Python Lesson 3

Miss Wallace

“Naming things with computers”

Lesson 2 Recap

- What did we do last week?

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- What did we do last week?
- Remember this question from our quiz last week? Let’s think about it again now we know how “print” statements work!

Question 1

Pick the option you think this code will print out

```
print("Hello!")
```

a. Hello!
b. Hi!
c. There will be an error and nothing will be printed out
d. print Hello!

Starter

Question: Think about what the word “variable” means to you. What sort of things are “variable”? What stays the same?

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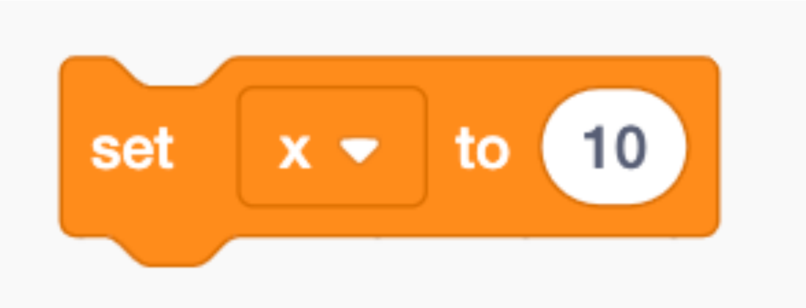

What we are building up to is modifying our conversation programs from last time so that we can actually “look” at the input users type in.

Lesson Aims

- I can consider the meaning of the word "variable" and how it applies to our own use of language and sentence structure
- I can understand that variables are used to store information in a program
- I can recognise examples of where variables may be used in modern applications

Activity 1 – Sentence Building

- You all have a post-it on your desk with X, Y or Z written on it

- Make sure you are arranged in a 3 so as a team you have one of each!

Activity 1 – Sentence Building

- You should write **on the back** of the post-it note a value to substitute in place of your letter in the following sentence:

“X went to the shop and bought Y and Z”

A couple rules:

X = something living who might go to the shop (a person, a celebrity, an animal, a mythical creature etc.)

Y = an object that they buy (add as many descriptive adjectives as you want)

Z = an object that they buy (add as many descriptive adjectives as you want)

Activity 1 – Sentence Building

- You have 5 minutes to write these on your post-it notes. Keep them hidden from your team mates!!

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- Write down on your sheet your sentence with the values substituted in.

Activity 1 – Sentence Building

- You have 5 minutes to write these on your post-it notes. Keep them hidden from your team mates!!
- Write down on your sheet your sentence with the values substituted in.
- Swap your X, Y and Z with the team next to you and write down this sentence with the new values substituted in.

Activity 1 – Sentence Building

- What can we learn from this?

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- We call X, Y and Z **variables** – we can substitute these for anything (within certain rules!) in our sentence and the sentence still (roughly) makes sense.

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- What can we learn from this?

- We call X, Y and Z **variables** – we can substitute these for anything (within certain rules!) in our sentence and the sentence still (roughly) makes sense.

- We use variables lots in programming. Let’s see how we can use variables to learn a person’s name.

Activity 2 - Variables

- We can make our conversations a bit more personal by keeping the person’s name as a variable.

```
name = input("What's your name?")
print("Nice to meet you " + name + "!")
```

- Does this feel more like a “proper” conversation?

Activity 2 - Variables

- We can make our conversations a bit more personal by keeping the person’s name as a variable.

```
name = input("What's your name?")
print("Nice to meet you " + name + "!")
```

- Does this feel more like a “proper” conversation?
- Try and implement a “name” variable so that the computer will say **your** name!

More variables

Tell me what will be printed out in each of these cases!

1)

```
animal = "dog"

print("You are a "+ animal)
```

More variables

Tell me what will be printed out in each of these cases!

2)

```python
animal = "dog"

animal = "cat"

print("You are a "+ animal)
```

More variables

Tell me what will be printed out in each of these cases!

3)

```
animal1 = "panda"

animal2 = "giraffe"

animal2 = animal1

print("Animal1 is a "+ animal1 + " and animal2 is a "+ animal2)
```

Challenge Question

```
animal1 = "koala"
animal2 = "kangaroo"
animal3 = "wombat"
animal4 = "platypus"
animal2 = animal4
animal1 = animal4
animal4 = "lemur"
animal2 = animal1
animal3 = animal2
```

Write down what each of the animals here will be!

Dealing with numbers

```
age = int(input("How old are you?"))
print("I'm " + age + " too!")
```

- We need to include the "int" so that the computer knows this variable represents a number!

Implementing your own variables

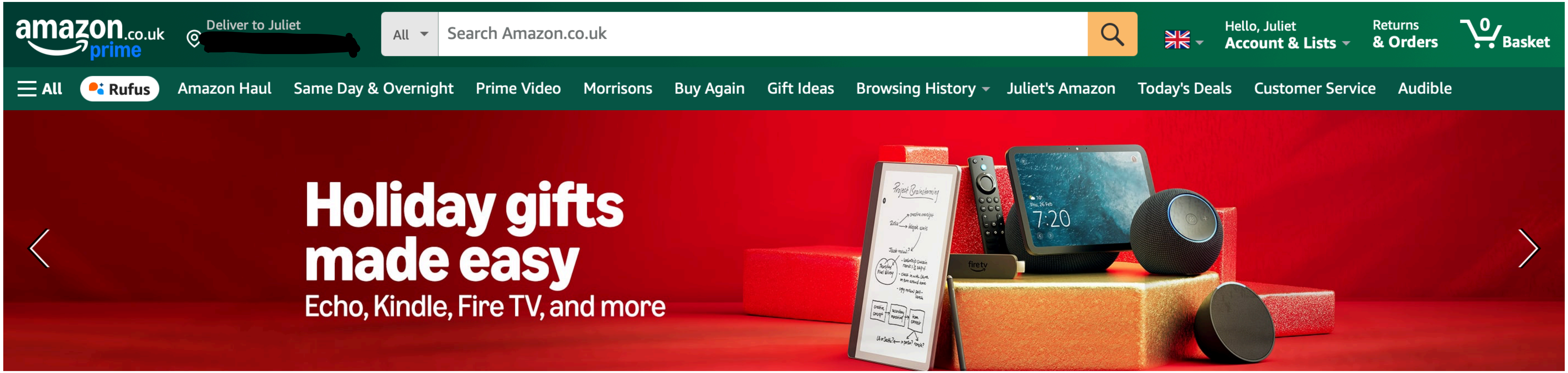
- Extend your conversation programs so you can implement 2 more variables. Maybe you could ask the other person’s favourite colour, or singer, or movie etc.
- If you ask for someone’s favourite colour, this could look like:

```
colour = input(“What’s your favourite colour?”)
print(“I like ” + colour + “ too!”)
```

- For a challenge, try including a variable that is a number in your conversations!

Real variables

(My first name is Juliet!)

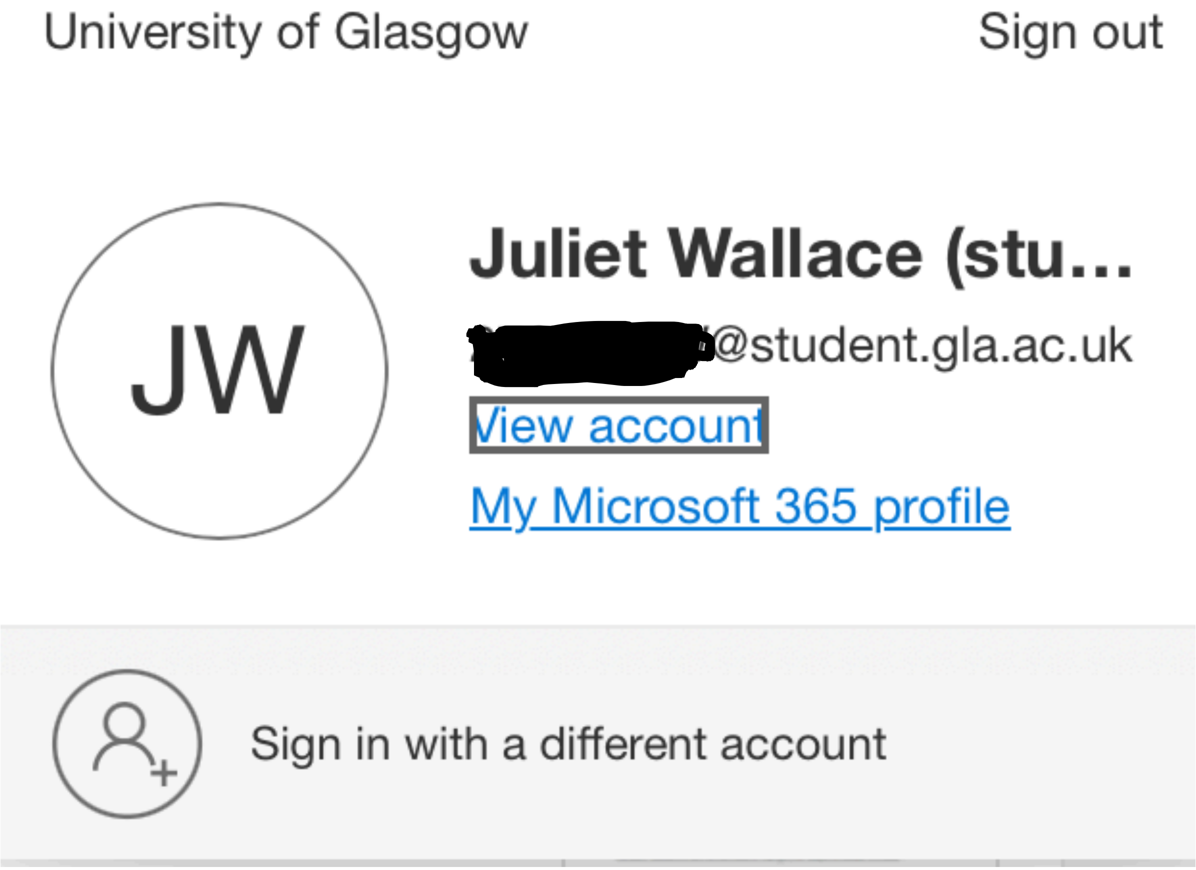


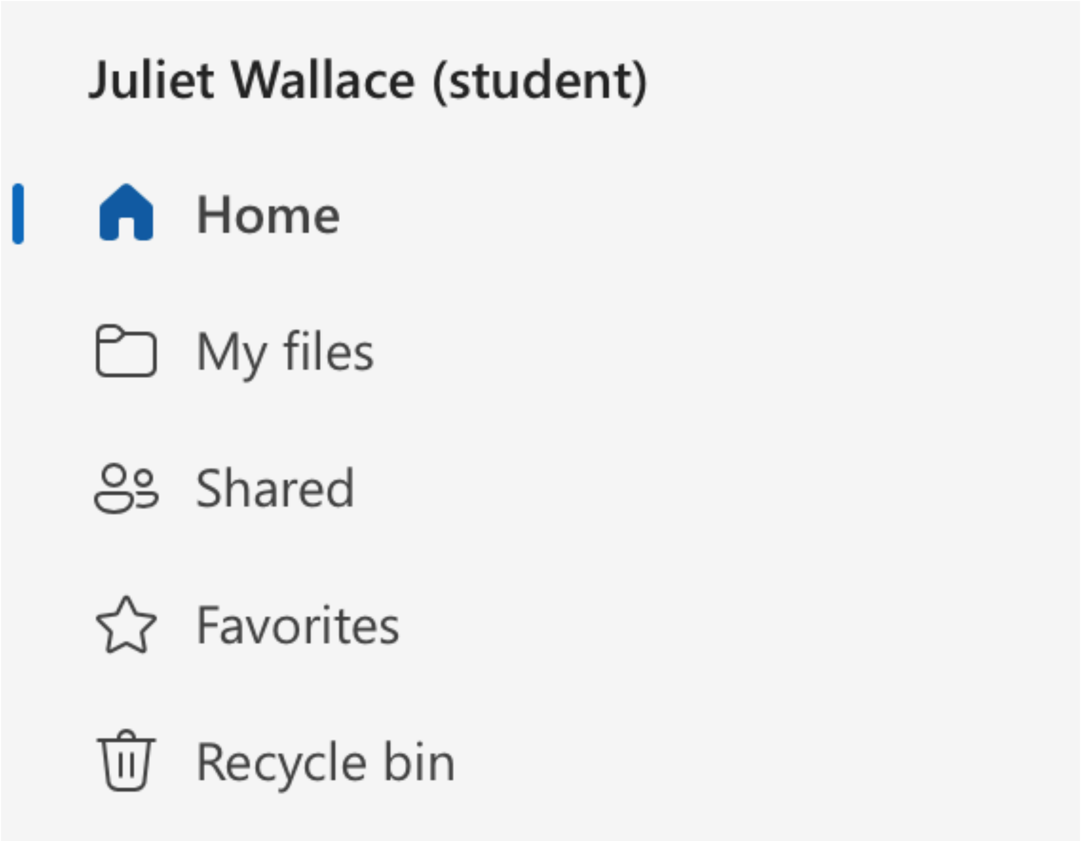
Let’s spot some variables here!

Image credit:
https://www.amazon.co.uk

Real variables

(My first name is Juliet!)





Can you find any other examples of this?

Challenge

- What variables can you spot here?

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What's the same and what's different?

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Remember this question....

A. x is equal to 10; y is equal to 5

B. x is equal to 5; y is equal to 5

C. x is equal to 10; y is equal to 10

D. x is equal to 5; y is equal to x

Can we answer this now?

How about this one....

Question 2

Pick the option you think this code will print out

```
x = 5
print(x)
```

a. x
b. 5
c. There will be an error and nothing will be printed out
d. =

Final discussion

- Does using variables make your conversations feel more personal? Does it feel closer to a human conversation? Is there anything still missing?